Ref. Ares(2019)6977887 - 11/11/2019

ESSnet Big Data II

Grant Agreement Number: 847375 — 2018-NL-BIGDATA

https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata
https://ec.europa.eu/eurostat/cros/content/essnetbigdata_en

Workpackage G

Financial transaction data

Deliverable 2: Conclusions on studies, and plan for year two

Final version, 8th November 2019

Workpackage country contact:

Valentin Chavdarov (BG, Bulgaria)
Christian Janz (DE, Germany)
Alessandra Righi (IT, Italy)
Johan Fosen (NO, Norway)
Pedro Campos (PT, Portugal)
Snežana Vrhovac (SI, Slovenia)

Workpackage Leader:

Johan Fosen (SSB, Norway)
johan.fosen@ssb.no
telephone : +47 990 17 864
mobile phone : +47 990 17 864

1. Background

The Essnet Big Data 2 work package "WPG Financial transactions" consists of a financial transaction data part (the main part) and a part on sharing economy platforms. Each of the two parts contains two steps according to the grant agreement:

1) Describing the financial transaction data infrastructure and metadata in each of the countries as well as identifying promising official statistics that can benefit from financial transaction data.
2) Performing empirical studies among promising official statistics.

For the financial transaction data part,

- Regarding step 1, a draft report of the findings is finished in 2019 as Deliverable 1.[1] A description of these findings will at the end of the project in autumn 2020 be updated in deliverable 4, the final report that will also contain the findings in the case studies.
- Step 2 is only relevant if financial transaction data can be accessed in time for performing the case study within the project period. Working on getting financial transaction is a work performed in parallel with the first step. Accessing financial transaction data has for most countries either been harder than expected, or it has taken longer time than expected. Still, some data are being accessed in time, cf. Section 2 below.

For the sharing economy part,

- step 1 should according to the grant agreement only be a literature study, and step 2 could only be performed after asking Eurostat. In mid-2019, it turned out that there were insufficient amount of literature to complete the task, since no one, neither within nor outside the ESS, seems to have been able to access sharing economy platform data. Consequently, it was agreed with Eurostat to alter the work on sharing economy, and further the originally planned step 2 of empirical studies on sharing economy data were even less relevant than initially planned. The conclusions on empirical studies in Section 2 is thus confined to the financial transaction part.

2. Conclusions of studies

The case studies of step 2 corresponds to Task 4 in the grant agreement, and its aim is to assess the statistics potential of a financial transaction data source. The potential usage of financial transaction data is of three types:

- Produce new official statistics or new indicators
- Replace some of the current data collection for the statistics.
- Adjusting the estimators of the production process using financial transaction data as auxiliary information.
- Quality assessment using financial transaction data as auxiliary information.

Produce new statistics or replacing the current data collection is the most ambitious use of financial transaction data and requires high quality of financial transaction data, whereas adjusting estimators puts somewhat less requirements on the data. However, if the quality is not sufficiently good for

[1] The sharing economy part of the project is also a part of Deliverable 1.

producing the official statistics based on the adjusted estimator, the adjusted estimator could still be valuable for assessing the quality, e.g. for bias estimation of the official statistics estimate.

Norway will perform a case study on the retail sales index, and this study has already been started. In addition, another case study might be performed.

The Italian case studies are still under evaluation because Istat has not received all the data yet. If data actually will allow it, Istat presumes to perform cases among the following three fields:

- pilot on the estimation of household expenditure of residents for tourism and Balance of payments flows
- pilot on use for forecasting of private consumption
- pilot on estimation e-commerce turnover by MCCs groups

Slovenia will perform a case study on the Retail sales index, but that depends on data being available. It is still too early to say whether data will be available.

Portugal has received financial transaction data from Tax Authority and will perform a case study. Information concerns all the formal transactions aggregated by vendor and buyer. There is the possibility of estimating statistics about hotel accommodation, transports and other sectors of interest.

Bulgaria will probably not receive financial transaction data and thus not be able to perform a case study.

Germany's amount of the WPG budget is too small to perform a case study.

3. Plan for year two

In the event of no financial transaction data accessed by a participating country, there cannot be a case study for this country. Also, the case study/studies possible to perform based on the actually accessed financial transaction data, might be very limited. At the time of writing the grant agreement, this meant that there was an uncertainty regarding the work of 2020 (year two). Therefore, it was included in the grant agreement that the plans for year two should be updated in the autumn of 2019. Below is the plan.

Since both Italy and Portugal will perform case studies in addition to Norway, these countries will in year two mostly work on case studies. Also, Slovenia might work on case studies. Germany's planned amount of work will be fulfilled by the completion of Deliverable 1 and in addition participating in the meetings and reviewing Deliverable 4. The remaining country with available time in year 2, is then only Bulgaria. As with Germany, Bulgaria will in year 2 participate in the meetings and working on reviewing Deliverable 4. In addition, Bulgaria will in year two define indicators based on FTD and include them in the National Statistical Program (NSP), thus identifying new promising statistical products based on FTD. Having NSP approved by government will ensure legal precondition to insist access to FTD. Knowing payment infrastructure is a must to define reasonable indicators based on FTD, and therefore Bulgaria will in year two continue to describe the infrastructure of the Bulgarian financial transaction data. The rest of the time will be allocated to the sharing economy part.

will in year two continue to describe the infrastructure of the Bulgarian financial transaction data, and the rest of the time will be allocated to the sharing economy part.

For the sharing economy part, as noted earlier the planned work has been adjusted to derive and analyse indicators of sharing economy related to official statistics, where these indicators are constructed based on financial transaction data. In deliverable 1, a methodological framework is laid out and the amount of relevant (aggregated) credit card data in each of the countries will be described. At the beginning of year 2, the results of this inventory will be discussed internally in WPG and with sharing economy-resources within Eurostat. Based on this discussion, it will be decided whether to move onto collecting aggregated credit card data from countries having such data, and in that case the indicators will be studied empirically. If there are insufficient amount of these data in the countries, the sharing economy part will proceed by studying the indicators non-empirically.